

1. Professional Profile

Muhammad Junayed is a final-year undergraduate student in Electronics and Telecommunication Engineering at Chittagong University of Engineering and Technology, commonly known as CUET, in Bangladesh.

His primary areas of interest include artificial intelligence, machine learning, computer vision, natural language processing, large language models, retrieval-augmented generation, cloud computing, backend engineering, MLOps, distributed systems, embedded systems, and the development of practical machine-learning applications.

His research experience includes convolutional neural networks, vision transformers, explainable artificial intelligence, Grad-CAM, transformer-based language models, educational NLP, medical-image analysis, semiconductor defect recognition, multimodal AI, and hallucination detection and mitigation in large language models.

He has contributed to three conference publications involving silicon wafer defect classification, breast-ultrasound image classification, and pedagogical evaluation of AI-powered tutors.

He is particularly interested in opportunities related to:

- AI engineering
- Machine-learning engineering
- Machine-learning research
- Computer vision
- Natural language processing
- Large language models
- Retrieval-augmented generation
- Cloud deployment
- Backend engineering
- MLOps
- Distributed software systems
- Applied AI research

2. Personal and Professional Information

- **Full name:** Muhammad Junayed
- **Location:** Chattogram, Bangladesh
- **Professional email:** mdjunayed573@gmail.com
- **University:** Chittagong University of Engineering and Technology
- **University abbreviation:** CUET
- **Department:** Electronics and Telecommunication Engineering
- **LinkedIn:** <https://www.linkedin.com/in/muhammad-junayed-ete20/>
- **GitHub:** <https://github.com/MD-Junayed000>
- **Google Scholar:** <https://scholar.google.com/citations?user=wObQzNsAAAAJ&hl=en>
- **ORCID:** <https://orcid.org/0009-0004-5445-9826>
- **Portfolio website:** <https://muhammadjunayed.vercel.app/>
- **Phone:** +880 1876220119

3. Education

3.1 Bachelor of Science in Electronics and Telecommunication Engineering

- **Degree:** Bachelor of Science in Electronics and Telecommunication Engineering
- **Institution:** Chittagong University of Engineering and Technology
- **Abbreviation:** CUET
- **Department:** Electronics and Telecommunication Engineering
- **Institution location:** Raozan, Chattogram, Bangladesh
- **Study period:** March 2022 to present
- **Current academic status:** Final-year undergraduate student
- **CGPA:** 3.51 out of 4.00
- **CGPA coverage:** Up to the seventh semester
- **Expected graduation date:** June, 2026

3.2 Higher Secondary Certificate

- **Qualification:** Higher Secondary Certificate

- **Abbreviation:** HSC
- **Institution:** Chattogram Cantonment Public School and College
- **Completion year:** 2020
- **Result:** GPA 5.00 out of 5.00

3.3 Secondary School Certificate

- **Qualification:** Secondary School Certificate
- **Abbreviation:** SSC
- **Institution:** Chattogram Cantonment Public School and College
- **Completion year:** 2018
- **Result:** GPA 5.00 out of 5.00

4. Undergraduate Thesis

4.1 Thesis Identity

- **Title:** Closing the Loop on RAG Hallucinations: Inference-Time Control via Dual Residual-Stream and FFN Activation Probes
- **Research area:** Large language models and retrieval-augmented generation
- **Status:** Ongoing undergraduate thesis
- **Supervisor:** Priyonti Paul Tumpa
- **Supervisor designation:** Assistant Professor
- **Supervisor department:** Department of Electronics and Telecommunication Engineering
- **Institution:** Chittagong University of Engineering and Technology

4.2 Research Problem

Large language models can generate responses that are fluent and convincing but factually unsupported, inconsistent with retrieved evidence, or disconnected from the supplied knowledge source. The thesis investigates methods for detecting and mitigating hallucinations during inference.

4.3 Proposed Research Direction

The proposed direction involves:

- Monitoring internal model states during generation
- Studying dual residual-stream representations
- Studying feed-forward network activation patterns
- Training or applying activation probes to identify hallucination risk
- Detecting signals associated with uncertainty or unsupported generation
- Using detected risk to control the retrieval and generation process
- Creating a closed-loop RAG process that can retrieve more evidence, revise an answer, or abstain when the available support is inadequate

4.4 Thesis Keywords

- Large language models
- LLM hallucination
- Hallucination detection
- Hallucination mitigation
- Retrieval-augmented generation
- Closed-loop RAG
- Dual residual stream
- Feed-forward network activations
- FFN activation probes
- Internal-state awareness
- Inference-time control
- Reliable AI
- Explainable AI
- Knowledge retrieval

4.5 Thesis Limitation in This Knowledge Source

No final experiment, thesis result, accuracy, benchmark, publication status, or completed implementation should be claimed unless those details are added to a future verified version of this document.

5. Work Experience

5.1 Software Engineer Intern - Poridhi.io

- **Organization:** Poridhi.io
- **Role:** Software Engineer Intern
- **Duration:** May 2025 to November 2025
- **Employment type:** Internship

The internship involved work related to:

- Software engineering
- Cloud computing
- Backend development

5.2 Industrial Trainee - Brain Station 23 PLC

- **Organization:** Brain Station 23 PLC
- **Role:** Industrial Trainee
- **Duration:** May 2025
- **Training type:** Industrial training

The training focused on:

- Modern software-engineering workflows
- Enterprise software systems
- Professional software-development practices

6. Research Interests

Muhammad Junayed's research and technical interests include:

- Artificial intelligence
- Machine learning
- Deep learning
- Computer vision
- Natural language processing
- Large language models
- Retrieval-augmented generation
- Agentic AI
- Vision transformers
- Transformer-based language models
- Explainable AI
- Grad-CAM
- AI-assisted education
- Medical-image analysis
- Semiconductor defect recognition
- Multimodal AI
- Cloud-based AI deployment
- Machine-learning operations
- Backend system development
- Distributed systems
- Internet of Things
- Embedded systems

7. Technical Skills

The following skills are listed in the source material. No numerical proficiency level is assigned unless explicitly stated.

7.1 Programming Languages

- Python
- C
- C++
- JavaScript
- PHP
- SQL

7.2 Frontend Development

- HTML
- CSS
- React
- Responsive user-interface development

7.3 Backend Development

- Node.js
- Express.js
- Flask
- FastAPI
- Django
- Django REST Framework
- REST API development
- Token-based authentication
- JSON Web Tokens
- WebSocket communication
- Real-time communication
- Socket.IO
- Celery

7.4 Machine Learning and Artificial Intelligence

- TensorFlow
- Keras
- scikit-learn
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Convolutional neural networks
- Vision transformers
- Graph neural networks
- Natural language processing
- Transformer-based language models
- Transfer learning
- Model evaluation
- Grad-CAM
- Explainable AI
- Classification
- Multiclass classification
- Imbalanced-data classification
- Retrieval-augmented generation
- Multimodal AI
- Optical character recognition
- CLIP-based vision processing

7.5 Language Models and RAG Technologies

- Ollama
- Llama 3.2
- LLaVA
- Hugging Face transformer models
- Pinecone vector search
- LangSmith tracing
- Model Context Protocol
- Document ingestion
- PDF processing
- OCR-based information extraction
- Vector-index updating
- Agentic workflows
- Web-search integration

- Conversation memory
- Session-history handling

7.6 Machine-Learning Operations

- MLflow
- ZenML
- Apache Airflow
- Experiment tracking
- Pipeline orchestration
- Model selection
- Reproducible machine-learning workflows
- Model deployment
- System monitoring concepts

7.7 Cloud and Infrastructure

- AWS fundamentals
- AWS EC2
- Docker
- Docker Compose
- Pulumi
- Infrastructure as Code
- Container networking
- Single-node cloud deployment
- Multi-node cloud deployment
- Nginx
- Git
- GitHub
- Linux

7.8 Databases and Data Platforms

- PostgreSQL
- MongoDB
- Redis
- BigQuery
- Pinecone
- Firebase

7.9 Messaging and Distributed Systems

- RabbitMQ
- Redis Pub/Sub
- Celery
- Temporal
- Socket.IO
- Dead-letter queues
- Event-driven architecture
- Asynchronous task processing
- Microservice architecture
- API gateway design
- Inter-service communication

7.10 Embedded Systems and IoT

- ESP32
- MQTT
- Firebase
- NVIDIA Jetson Nano
- RFID systems
- JT308 RFID module
- Camera-based recognition systems
- IoT automation
- Energy monitoring

- Remote AC load control

7.11 Engineering, Analysis, and Productivity Software

- MATLAB
- Visual Studio Code
- Cisco Packet Tracer
- Jupyter Notebook
- LaTeX
- Microsoft Word
- Microsoft PowerPoint
- Power BI
- Adobe Photoshop
- DaVinci Resolve

8. Relevant Coursework

8.1 Electronics and Communication

- Electronic Devices and Circuits
- Electronics I
- Electronics II
- Digital Logic Design
- Analog Communication
- Digital Communication
- Wireless Communication
- Microwave and Antenna Engineering
- Optical Fiber Communication
- Radar and Satellite Communication

8.2 Signal Processing and Control

- Signals and Systems
- Digital Signal Processing
- Control Systems Engineering

8.3 Computing

- C Programming
- C++ Programming
- Object-Oriented Programming
- Data Structures and Algorithms
- Microprocessors and Microcontrollers
- Web Development Fundamentals
- HTML
- CSS
- JavaScript

8.4 Mathematics

- Calculus
- Ordinary Differential Equations
- Partial Differential Equations
- Linear Algebra
- Matrix Analysis
- Vector Analysis
- Complex Variables
- Probability
- Statistics

9. Research and Publications

9.1 Enhancing Defect Recognition: Convolutional Neural Networks for Silicon Wafer Map Analysis

- **Document ID:** publication_wafer_map_cnn

- **Authors:** Muhammad Junayed, Tanzeem Tahmeed Reza, and Md. Saiful Islam
- **Muhammad Junayed's author position:** First author
- **Conference:** 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering
- **Conference abbreviation:** ICAEEE 2024
- **Conference location:** Gazipur, Bangladesh
- **Publisher:** IEEE
- **Publication date:** April 25, 2024
- **Pages:** 1-6
- **DOI:** 10.1109/ICAEEE62219.2024.10561853
- **DOI link:** <https://doi.org/10.1109/ICAEEE62219.2024.10561853>
- **Code repository:** Not provided

Research Problem

Manual identification of silicon wafer-map failure patterns can be time-consuming and inefficient. The study investigated an automated method for multiclass silicon wafer defect recognition.

Dataset

The study used a combined dataset from which 902 images were selected through random sampling across the target classes.

The following information is not documented in the provided sources:

- Exact public dataset name
- Original dataset size
- Complete class distribution
- Exact train-validation-test split

Approach

A deep convolutional neural network was developed for multiclass wafer-map classification. The proposed model was compared with established deep-learning architectures, including:

- VGGNet
- ResNet
- EfficientNet

Gradient-weighted Class Activation Mapping, or Grad-CAM, was used to identify image regions that contributed to model predictions.

Main Methods

- Convolutional neural network
- Multiclass image classification
- Baseline-model comparison
- Grad-CAM visualization
- Accuracy evaluation
- F1-score evaluation

Main Result

The proposed convolutional neural network achieved an accuracy of 91.7 percent.

Evaluation Limitation

Exact per-class precision, recall, F1-score, confusion-matrix values, and complete baseline results are not included in the provided sources.

Keywords

Silicon wafer maps; semiconductor manufacturing; defect recognition; CNN; deep learning; Grad-CAM; image classification; explainable AI.

9.2 SmolLab_SEU at BEA 2025 Shared Task: A Transformer-Based Framework for Multi-Track Pedagogical Evaluation of AI-Powered Tutors

- **Document ID:** publication_bea_2025
- **Authors:** Md. Abdur Rahman, Md. Al Amin, Sabik Aftahee, Muhammad Junayed, and Md. Ashiqur Rahman
- **Muhammad Junayed's author position:** Fourth author
- **Team:** SmolLab_SEU

- **Venue:** Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications
- **Workshop abbreviation:** BEA 2025
- **Publisher:** Association for Computational Linguistics
- **Publication date:** July 2025
- **Pages:** 1127-1134
- **Official paper link:** <https://aclanthology.org/anthology-files/pdf/bea/2025.bea-1.88.pdf>
- **DOI:** Not provided
- **Code repository:** Not provided

Research Problem

The study addressed the automatic evaluation of the pedagogical abilities of AI-powered tutors. The objective was to assess tutor responses across multiple educational dimensions.

Dataset

The research used the MRBench dataset, which contains mathematical tutoring dialogues.

Evaluated Tracks

1. Mistake identification
2. Mistake location
3. Providing guidance
4. Actionability
5. Tutor identity prediction

Models Evaluated

- DeBERTa-V3
- RoBERTa-Large
- SciBERT
- EduBERT

Approach

The transformer models were strategically fine-tuned for the individual tracks. Weighted loss functions were used to reduce the effect of class imbalance.

Shared-Task Rankings

- Tutor Identity: 9th place
- Actionability: 16th place
- Providing Guidance: 19th place
- Mistake Identification: 20th place
- Mistake Location: 22nd place

Exact F1 Results

- Tutor Identity: 0.8621
- Actionability: 0.6284
- Providing Guidance: 0.4933
- Mistake Identification: 0.6617
- Mistake Location: 0.4935

Main Finding

DeBERTa-based architectures generally achieved better performance than the other evaluated transformer models across the shared-task tracks.

Keywords

Educational NLP; AI-powered tutors; pedagogical evaluation; transformer models; DeBERTa; RoBERTa; SciBERT; EduBERT; class imbalance; shared task; MRBench.

9.3 Leveraging Vision Transformer to Classify Cancer Stages from Breast Ultrasound Images

- **Document ID:** publication_breast_ultrasound_vit
- **Authors:** Muhammad Junayed and Tanzeem Tahmeed Reza
- **Muhammad Junayed's author position:** First author

- **Conference:** 2025 IEEE International Conference on Signal Processing, Information, Communication and Systems
- **Conference abbreviation:** SPICSCON 2025
- **Conference location:** Rajshahi University, Bangladesh
- **Publisher:** IEEE
- **Publication date:** November 21, 2025
- **Pages:** 577-580
- **DOI:** 10.1109/SPICSCON69221.2025.11504239
- **DOI link:** <https://doi.org/10.1109/SPICSCON69221.2025.11504239>
- **Code repository:** Not provided

Research Problem

Breast-ultrasound interpretation can be affected by operator dependence and imaging limitations. The study investigated deep-learning-based classification of breast-ultrasound images.

Dataset

- **Dataset:** Breast Ultrasound Images Dataset
- **Common abbreviation:** BUSI
- **Target classes:** Normal, benign, and malignant

Approach

A tailored Vision Transformer-based encoder was developed for multiclass breast-ultrasound classification. The proposed model was compared with a standard Vision Transformer baseline.

Main Methods

- Vision Transformer
- Medical-image classification
- Multiclass classification
- Transformer-based image encoding
- Baseline-model comparison

Evaluation Information Not Provided

The following values are not included in the source material:

- Final accuracy
- Precision
- Recall
- F1-score
- AUC
- Confusion matrix
- Dataset split
- Baseline comparison values

These values should be added only after verification from the published paper.

Keywords

Breast cancer; breast ultrasound; BUSI dataset; Vision Transformer; medical-image analysis; deep learning; multiclass classification.

10. Selected Projects

10.1 AroBot - Agentic Multimodal RAG Chatbot System

- **Document ID:** project_arobot
- **Project type:** Artificial intelligence, multimodal RAG, and backend system
- **Role:** Developer
- **Canonical status:** Functional prototype with production-style components; public production and clinical deployment are not verified
- **Repository:** Not provided
- **Demonstration:** <https://muhammadjunayed.vercel.app/projects/arobot>

Problem

The project was designed to process medical-related text, images, prescriptions, and PDF documents through an AI-assisted conversational interface.

Technologies

- Ollama
- Llama 3.2 3B
- LLaVA 7B
- Pinecone
- FastAPI
- Optical character recognition
- CLIP
- LangSmith
- DuckDuckGo search
- Vector databases
- Rate-limited APIs
- Responsive web interface

Functional Components

- Text processing
- Image processing
- PDF processing
- Prescription OCR
- Retrieval-augmented generation
- Pinecone vector retrieval
- Web-search integration
- Conversation-memory management
- Session-history recording
- Vector-index updating
- LangSmith tracing and observability
- Auditable session histories
- Health-metric handling as described in the CV, although the exact metric definitions are not documented

Implementation Context

Ollama was used to run local language models. Llama 3.2 3B supported text-oriented generation, while LLaVA 7B supported multimodal interaction. A CLIP-based vision pipeline was used for image-related processing. Pinecone was used for vector retrieval, FastAPI was used for the backend, LangSmith was used for tracing, and DuckDuckGo search was used for web-search integration.

Outcome

The project produced a multimodal medical-information assistant capable of handling text, images, PDFs, OCR, retrieval, conversational memory, and session history.

Evaluation and Safety Limitations

The following are not documented:

- Clinical validation
- Medical-expert evaluation
- Retrieval precision
- Retrieval recall
- Faithfulness score
- Hallucination rate
- Response latency
- Concurrent-user capacity
- Security testing
- Production-traffic statistics

Keywords

Agentic RAG; multimodal chatbot; medical-information assistant; Ollama; Llama 3.2; LLaVA; Pinecone; FastAPI; OCR; CLIP; LangSmith; Model Context Protocol.

10.2 Uber Fare Prediction with MLOps Orchestration

- **Document ID:** project_uber_fare_mlops
- **Project type:** Machine learning and MLOps
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/MLOPS_Uber_fare
- **Deployment framework:** Flask

Problem

The project addressed fare prediction while establishing a reproducible machine-learning pipeline for training, model selection, experiment tracking, deployment, and monitoring.

Technologies

- Apache Airflow
- ZenML
- MLflow
- Flask
- PostgreSQL
- Ensemble machine-learning models

Approach

- Airflow was used for workflow scheduling.
- ZenML was used for pipeline orchestration.
- MLflow was used for experiment tracking and model comparison.
- Flask was used to expose the selected model.
- PostgreSQL was used for metadata storage and system-level logging.

Outcome

An end-to-end MLOps pipeline was created to support experimentation, model selection, reproducibility, deployment, and monitoring.

Keywords

Uber fare prediction; MLOps; Airflow; ZenML; MLflow; Flask; PostgreSQL; ensemble learning.

10.3 Bangladesh Medicine Scraper

- **Document ID:** project_bangladesh_medicine_scrape
- **Project type:** Web scraping, backend development, and data engineering
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/Bangladesh_Medicine_Scraper

Problem

The project aimed to collect structured medicine information from MedEx and expose the collected data through a manageable backend system.

Technologies

- Django
- Django REST Framework
- Scrapy
- Playwright
- PostgreSQL
- Browser-cookie reuse
- Token-authenticated REST APIs
- Django Admin

Approach

A Scrapy and Playwright pipeline reused verified browser cookies to crawl medicine-related information. The extracted data was stored in PostgreSQL and exposed through Django Admin and authenticated REST APIs.

Outcome

The project produced a medicine-data collection and API-delivery pipeline.

Keywords

Medicine scraper; MedEx; Django; Django REST Framework; Scrapy; Playwright; PostgreSQL; REST API.

10.4 NodeScape - Graph Algorithm Visualizer and Graph-Type Predictor

- **Document ID:** project_nodescape
- **Project type:** Full-stack graph learning and educational visualization
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/NodeScape-Low_Bias_High_Variance

Problem

The project combined graph-algorithm education with machine-learning-based graph-type prediction.

Technologies

- React
- Flask
- TensorFlow
- LightGIN graph neural network
- Docker

Features

- Interactive graph construction and visualization
- Step-by-step breadth-first search visualization
- Step-by-step depth-first search visualization
- Graph-type prediction
- Docker deployment support

Outcome

A full-stack graph visualization and graph-classification platform was developed.

Keywords

Graph visualization; BFS; DFS; graph classification; LightGIN; graph neural network; React; Flask; TensorFlow.

10.5 Bistro-92 Restaurant Management System

- **Document ID:** project_bistro_92
- **Project type:** Microservice-based full-stack system
- **Role:** Developer
- **Repository:** <https://github.com/MD-Junayed000/Bistro-92>

Problem

The project aimed to develop a scalable restaurant-management platform using independently coordinated backend services.

Technologies

- Go
- React
- RabbitMQ
- PostgreSQL
- Redis
- Temporal
- Docker

Architecture

The project used a Dockerized microservice architecture with asynchronous communication, persistent storage, caching, and workflow coordination.

Outcome

A restaurant-management platform was implemented using microservices, messaging, caching, persistent storage, and Temporal workflow orchestration.

Keywords

Restaurant management; microservices; Go; React; RabbitMQ; PostgreSQL; Redis; Temporal; Docker.

10.6 Face Recognition and RFID Smart Attendance System

- **Document ID:** project_smart_attendance
- **Project type:** Computer vision, embedded AI, and IoT
- **Role:** Developer
- **Repository:** <https://github.com/MD-Junayed000/Smart-Attendance-System>

Problem

The project addressed real-time attendance verification using multiple identity-verification mechanisms.

Hardware

- NVIDIA Jetson Nano
- JT308 RFID module
- Canon EOS 250D camera

Technologies and Models

- MTCNN for face detection
- ResNet-18 for face recognition
- RFID-based identification
- Docker

Approach

The system combined face detection, face recognition, and RFID verification to support multimodal attendance tracking.

Outcome

A Dockerized real-time attendance system was developed with multimodal identity verification.

Keywords

Smart attendance; face recognition; RFID; Jetson Nano; MTCNN; ResNet-18; multimodal verification; Docker.

10.7 Credit Card Fraud Detection

- **Document ID:** project_credit_card_fraud
- **Project type:** Machine learning and imbalanced classification
- **Role:** Developer
- **Repository:** <https://github.com/MD-Junayed000/Credit-Card-Fraud-Detection-App>
- **Deployment framework:** Streamlit

Problem

The project addressed fraud detection in highly imbalanced transaction data.

Models

- Random Forest
- XGBoost
- LightGBM

Reported Results

- Approximate AUPRC: 0.98
- Approximate F1-score: 0.96

Outcome

The trained fraud-detection model was deployed through Streamlit.

Keywords

Credit-card fraud; imbalanced classification; Random Forest; XGBoost; LightGBM; AUPRC; F1-score; Streamlit.

10.8 EEG Data Analysis and Alcoholism Detection

- **Document ID:** project_eeg_alcoholism
- **Project type:** Biomedical signal analysis and machine learning
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/EEG_Signal_Data_Analysis

Problem

The project investigated classification of EEG signals for alcoholism detection.

Models

- K-nearest neighbors
- Support vector machine

Reported Result

The project reports accuracy of up to 99 percent..

Keywords

EEG; alcoholism detection; biomedical signal processing; KNN; SVM; classification.

10.9 Asynchronous Task Processing System with Flask, Celery, and AWS Pulumi

- **Document ID:** project_flask_async
- **Project type:** Backend engineering, cloud computing, and distributed systems
- **Role:** Developer
- **Repository:** <https://github.com/MD-Junayed000/Async-tasks-main>

Technologies

- Flask
- Celery
- RabbitMQ
- Redis
- Docker
- AWS EC2
- Pulumi

Approach

The project implemented distributed asynchronous task-processing pipelines. Flask provided the application backend, Celery managed background tasks, RabbitMQ acted as the task broker, Redis supported task-state handling, Docker provided containerization, and Pulumi automated AWS infrastructure provisioning.

Deployment

- Single-node AWS EC2 deployment
- Multi-node AWS EC2 deployment
- Distributed worker execution

Outcome

The project demonstrated asynchronous job processing, containerized deployment, infrastructure provisioning, and distributed worker deployment.

Keywords

Asynchronous processing; Flask; Celery; RabbitMQ; Redis; Docker; AWS EC2; Pulumi; Infrastructure as Code.

10.10 Node.js Asynchronous Task Processing System

- **Document ID:** project_node_async
- **Project type:** Distributed backend system
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/Async_task_node.js-main

Technologies

- Node.js
- RabbitMQ

- Redis
- Socket.IO
- Docker Compose
- Pulumi

Architecture

- RabbitMQ was used for job queueing.
- Redis Pub/Sub was used for job-status updates.
- Socket.IO provided real-time browser feedback.
- Dead-letter queues supported failed-message recovery.
- Docker Compose managed the services.
- Worker instances could be scaled through a command such as `--scale worker=n`.

Outcome

A scalable asynchronous task-processing architecture was developed with live task tracking, failure recovery, and horizontally scalable workers.

Keywords

Node.js; asynchronous tasks; RabbitMQ; Redis Pub/Sub; Socket.IO; dead-letter queue; Docker Compose; Pulumi.

10.11 RickshawX - Smart Mobility for CUET

- **Document ID:** project_rickshawx
- **Project type:** Full-stack microservice ride-sharing platform
- **Role:** Developer
- **Target environment:** CUET campus
- **Repository:** https://github.com/MD-Junayed000/RickshawX-Smart_Mobility_for_CUET

Problem

The project aimed to provide a smart ride-sharing and trip-management platform for the CUET community.

Technologies

- Node.js
- Express.js
- React
- MongoDB
- Redis
- RabbitMQ
- Docker
- Docker Compose
- Nginx
- JSON Web Tokens

Services

- Authentication service
- Ride service
- Trip service
- Payment service
- Notification service

Features and Architecture

- JWT authentication
- Centralized API gateway
- Asynchronous inter-service communication
- RabbitMQ-based event processing
- GPS-based fare calculation
- Haversine-distance calculation
- Redis caching
- Trip management
- Ratings

- Real-time notifications
- Nginx routing or reverse-proxy support
- Docker-based orchestration

Outcome

A production-style microservice platform was implemented for campus ride sharing.

Keywords

Ride sharing; CUET; microservices; Node.js; Express; React; MongoDB; Redis; RabbitMQ; Nginx; Docker; Haversine formula.

10.12 Aroma Pharmacy E-Commerce Platform

- **Document ID:** project_aroma_pharmacy
- **Project type:** E-commerce web development
- **Role:** Developer
- **Repository:** <https://github.com/MD-Junayed000/Aroma-Pharmacy>

Problem

The project aimed to create an online pharmaceutical product-browsing and e-commerce platform.

Technologies

- HTML
- CSS
- PHP
- JavaScript

Outcome

A pharmaceutical e-commerce platform was developed.

Keywords

Pharmacy e-commerce; PHP; HTML; CSS; JavaScript; product browsing.

10.13 Home Power Automation System

- **Document ID:** project_home_power_automation
- **Project type:** IoT and embedded automation
- **Role:** Developer
- **Repository:** https://github.com/MD-Junayed000/Home_Power_Automation

Problem

The project addressed prepaid energy monitoring and remote control of AC electrical loads.

Technologies

- ESP32
- MQTT
- Firebase
- Energy-monitoring sensors
- Remote AC load-control components

Features

- Prepaid energy monitoring
- Remote AC load control
- MQTT-based communication
- Firebase-based data handling

Outcome

An ESP32-based IoT power-monitoring and automation system was developed.

Keywords

Home automation; power monitoring; prepaid energy; ESP32; MQTT; Firebase; AC load control; IoT.

10.14 DataBench Question Answering System

- **Document ID:** project_databench_qa
- **Project type:** Natural language processing and tabular question answering
- **Competition:** SemEval 2024 Task 8
- **Role:** Developer and competition participant
- **Repository:** <https://github.com/MD-Junayed000/Tabular-QA>

Problem

The project addressed question answering over structured tabular data.

Technologies and Methods

- Rule-based information extraction
- Semantic-similarity matching
- Transformer-based natural language processing
- BERT-large-uncased-SQuAD2

Approach

The system combined deterministic extraction, semantic matching, and transformer-based question answering to generate data-driven responses from structured tables.

Reported Outcome

The project was submitted to SemEval 2024 Task 8: DataBench and is described in the source CV as a finalist project.

Keywords

DataBench; SemEval 2024; tabular question answering; BERT; semantic similarity; information extraction; NLP.

11. Extracurricular Activities and Leadership

11.1 Televerse 1.0, CUET

- **Role:** Team Lead and Organizer
- **Duration:** October 2025 to November 21, 2025
- **Event:** Televerse 1.0
- **Institution:** CUET

Responsibilities

- Led the IOTrix competition
- Coordinated interdisciplinary teams
- Supported work involving IoT, automation, hardware, software, and embedded-system prototypes
- Contributed to event organization and execution

IOTrix is Bangladesh's first Shark Tank-style hardware-software integrated automation competition.

11.2 CUET Photographic Society

- **Organization:** CUET Photographic Society
- **Abbreviation:** CUETPS
- **Role:** Assistant Design Secretary
- **Duration:** September 23, 2024 to March 11, 2026

Responsibilities

- Contributed to organizational branding
- Created visual-design materials
- Supported media-content production
- Contributed to campaigns and events

11.3 3rd International Conference on Electrical, Computer and Communication Engineering

- **Conference:** 3rd International Conference on Electrical, Computer and Communication Engineering
- **Abbreviation:** ECCE
- **Host institution:** CUET

- **Role:** Event Manager
- **Duration:** February 23-25, 2023

Responsibilities

- Supported event coordination
- Managed logistics
- Contributed to conference execution
- Assisted with an international engineering conference hosted at CUET

12. Awards and Achievements

12.1 ETE Infixon Case Solving Competition

- **Achievement:** Champion
- **Institution:** CUET
- **Category:** Case-solving competition

12.2 ETE Programming Hackathon

- **Achievement:** First runner-up
- **Organizer:** Department of Electronics and Telecommunication Engineering, CUET
- **Category:** Programming competition

12.3 Revealing the Unseen: Season VI

- **Achievement:** Finalist
- **Organizer:** CUET Photographic Society
- **Category:** Photography competition

12.4 BitFest Datathon 2025

- **Achievement:** 13th place out of 108 teams
- **Event:** BitFest Datathon
- **Organizer:** Department of Computer Science and Engineering, Khulna University of Engineering and Technology
- **Year:** 2025
- **Category:** Data science competition

12.5 SemEval 2024 Task 8: DataBench

- **Achievement:** Finalist, as reported in the source CV
- **Category:** Natural-language processing and tabular question answering

12.6 Techathon: The Heart of Robolution 1.0

- **Achievement:** Third place
- **Organizer:** Department of Mechatronics and Industrial Engineering, CUET
- **Category:** Technology competition

12.7 BEA 2025 Shared Task

- **Achievement:** Finalist and top-10 result in one track, as reported in the source CV
- **Event:** BEA 2025 Shared Task on Pedagogical Evaluation of AI Tutors
- **Organizer:** ACL SIGEDU
- **Best rank:** 9th place in Tutor Identity
- **Best reported score:** Exact F1 of 0.8621
- **Additional participation:** Competitive rankings in four additional tracks

12.8 Break the Monolith Microservice Hackathon

- **Achievement:** Honorable Mention
- **Organizer:** IEEE Computer Society, CUET
- **Category:** Microservices and software architecture

13. Languages

13.1 Bangla

- **Proficiency:** Native

13.2 English

- **Proficiency:** Working and academic proficiency
- **Evidence:** Conference-paper writing, research documentation, technical documentation, project presentations, and software-development communication
- **Standardized test score:** Not provided